

VineVision: Mobile Multi-Camera Yield Forecasting for Mid-Size Greenhouse Tomato Production

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Abstract – Ontario's greenhouse tomato sector produces 97.6% of Canada's greenhouse tomato output, yet yield forecasting remains largely manual, with error rates of 10–20% putting mid-size operations at risk of \$120,000 - \$150,000 in losses per season. VineVision addresses this gap through a mobile, infrastructure-light crop monitoring system consisting of a custom cart and mast assembly, a four-camera RGB and depth sensing array, and a cloud-hosted microservice software pipeline. The system continuously monitors fruit count, ripeness, size, estimated weight, and flower pollination state, delivering zone-level yield predictions up to 8 weeks before harvest through a web-based dashboard. Two machine learning models were developed and validated: a Tomato Ripeness Model (mAP@50: 0.947) and a Flower Pollination Model (mAP@50: 0.902), both surpassing the 90% mean average precision target. Weight estimation was validated against ground truth measurements across 30 tomatoes, with 90% of results falling within 5% of actual weight. VineVision requires no custom infrastructure modification, making it accessible to mid-size growers who have been underserved by existing solutions.

Keywords – Yield forecasting, computer vision, precision agriculture, greenhouse automation, depth sensing

I. INTRODUCTION

Modern greenhouse agriculture enables year-round, high-value crop production in tightly controlled environments. Ontario leads this sector in Canada, producing 97.6% of the country's 294 million kilograms of greenhouse tomatoes in 2022 and contributing to \$5.0 billion in national greenhouse sales in 2024 [1][2]. Despite this scale, yield forecasting remains largely manual, producing error rates of 10–20% that propagate across the supply chain, driving overproduction, supply mismatches, and food waste costing the global economy USD \$1 trillion annually [3][4].

Existing automated solutions, namely IUNU's LUNA AI platform, require custom aluminium rail infrastructure installed at the greenhouse truss level, limiting adoption to large-scale operations [5]. Academic research has demonstrated that multi-camera depth fusion achieves fruit detection accuracies above 91% and meaningfully improves weight estimation over single-camera approaches [6][7], but these methods have not been translated into accessible commercial tools for mid-size growers. VineVision addresses this gap by applying depth-fused multi-camera computer vision in a mobile, low-infrastructure form factor suited to the operational realities of Ontario's greenhouse sector.

II. PROJECT MOTIVATION, SCOPE AND OBJECTIVES

For a mid-size Canadian greenhouse operation, manual forecasting errors of 10–20% put an estimated \$120,000–\$150,000 at risk each growing season [3]. Growers absorb this uncertainty by overproducing, distributors face unreliable supply commitments, and consumers experience inconsistent pricing, all avoidable consequences of insufficient crop visibility. No affordable, infrastructure-light solution currently exists for this market segment.

VineVision was developed to provide accurate, automated yield forecasts weeks before harvest without requiring custom greenhouse modifications. The system is scoped to tomato crops in mid-size Ontario greenhouse operations, targeting the greenhouse growers, the primary user, while addressing downstream impacts on distributors and retailers. The core objective is to reduce forecast error below the 10% threshold through continuous monitoring of fruit count, growth stage, and ripeness using a four-camera depth-sensing array mounted on a mobile cart.

The need driving this project can be stated as follows: *Ontario greenhouse tomato growers require an affordable, non-invasive yield forecasting system that delivers actionable harvest predictions weeks in advance, without demanding infrastructure investment beyond the reach of mid-size operations.*

III. SUMMARY OF ENGINEERING ANALYSIS AND DESIGN METHODS

Following concept generation, engineering specification development, and initial prototype design completed in SYDE 461, using techniques such as QFD, user interviews, and decision matrices, the subsections below outline how each subsystem was further developed and refined in SYDE 462. Updates to user requirements, field visit findings, and iterative testing informed revisions across the hardware, mechanical, machine learning, and software platform subsystems.

A. HARDWARE DESIGN

Hardware design was guided by five requirements: no greenhouse infrastructure modification, sensing capability for fruit detection and sizing, affordability for mid-size growers, reliable operation in a humid environment, and compatibility with a mobile single-operator workflow.

Sensing architecture and camera arrangement – The final sensing stack combined one Intel RealSense D435 with three Raspberry Pi Camera Module 3 units. This configuration was selected through concept comparison supported by a weighted decision matrix that considered sensing capability, integration complexity, synchronization feasibility, cost, and suitability for mobile greenhouse deployment. The D435 was selected as the primary centre sensor because it provides registered RGB-depth output needed for short-range tomato size and weight estimation. The Raspberry Pi Camera Module 3 units were selected for the auxiliary views because they provide high-resolution RGB imaging at low cost with native CSI connectivity. Camera arrangement was informed by geometric coverage analysis and preliminary occlusion testing, which were used to compare single-view and multi-view layouts before fixing the final geometry. The final arrangement used one forward-facing depth camera at 0°, two side RGB cameras, and one under center RGB camera. At a nominal 0.5 m standoff, the side views increased lateral coverage by approximately 0.25–0.30 m relative to the centre view while maintaining sufficient overlap for multi-view calibration.

Compute configuration, power system, and synchronization - A distributed architecture using one Raspberry Pi 4 Model B and three Raspberry Pi Zero 2 W units was selected over a centralized single-board configuration through weighted comparison of cost, wiring complexity, modularity, and expected processing load. The Pi 4 was assigned to RealSense ingestion, system coordination, and backend communication because these tasks imposed the greatest bandwidth and compute demand. Each Pi Zero 2 W was assigned one auxiliary camera, eliminating the need for a multi-camera adapter and improving modularity. Synchronization strategy was selected through comparison of hardware-triggered, sequential, and software-timestamped capture approaches, followed by bench testing of timing performance and implementation complexity. The final system used software-triggered acquisition with a shared future timestamp, which minimized wiring complexity

while maintaining adequate timing performance. The power system was sized using component-level power draw and an 8-hour operating requirement. Under representative operation, the system draws 13.4–15.4 W, so a 20 W design ceiling was adopted as summarized in Appendix B.2. For an 8-hour deployment, this corresponded to 160 Wh required energy; with 80% usable depth-of-discharge, a 200 Wh battery was selected. Total hardware cost was CAD 485, within the CAD 500 target.

Table 1: Hardware User Requirements and Engineering Specifications

User Requirement	Engineering Specification	Final Design Response
Multi-view sensing for fruit detection and sizing	≥ 1 depth sensor + auxiliary RGB views	1 $\times$ Intel RealSense D435 + 3 $\times$ RPi Camera Module 3
Reduce occlusion while preserving overlap	Side-view coverage with shared overlap region	Centre depth view + RGB side views + under center RGB
Low-cost architecture	Total hardware cost < CAD 500	Final BOM cost: CAD 485
Mobile and low-power	Average draw ≤ 20 W, battery-operable	13.4–15.4 W measured, 200 Wh battery selected
Reliable synchronized capture	Multi-node capture skew < 100 ms	Software-triggered capture with < 80 ms skew

B. MECHANICAL DESIGN

Mechanical design was guided by five requirements: adjustable sensing height across the fruiting zone, stable relative camera geometry during motion and capture, resistance to humidity and splash exposure, modular construction for maintenance and iteration, and compatibility with narrow greenhouse rows.

Mast design and adjustment – A telescoping single-mast architecture was selected over fixed-height and wide multi-arm concepts using a weighted decision matrix informed by grower feedback and field measurements. Evaluation criteria included adjustability across crop growth stages, row compatibility, stability, ease of use, and fabrication complexity. Fixed-height designs were incompatible with the observed 0.4–0.6 m shift in fruiting height over the growth cycle [8], while wide multi-arm concepts increased cart width and interfered with row clearance. The selected mast provided 0.8–2.2 m vertical adjustment and used an intuitive lock mechanism to support tool-free single-operator repositioning. A single-mast architecture also minimized cantilevered mass and simplified alignment of all camera modules to a common rigid frame

Environmental protection – Environmental protection requirements were defined from field observations of humidity, condensation risk, and intermittent splash exposure, then translated into minimum enclosure specifications through the engineering requirements process. IP54 was defined as the minimum acceptable protection level, and enclosure selection was finalized by comparing available options against sealing performance, durability, and integration constraints. The final design used an IP54-rated enclosure with sealed cable glands to provide margin above this requirement. Aluminium, PETG, and stainless steel were used for structural components and mounts to reduce corrosion risk and maintain dimensional stability in humid conditions.

Camera geometry, modularity, and calibration – Camera mounting and modularity were developed through iterative prototyping, with bracket concepts evaluated for adjustability, rigidity, serviceability, and preservation of calibration. The final camera layout used independent modular brackets attached to a common mast extrusion. This allowed rapid adjustment and replacement of individual camera modules without changing the rest of the system. The side-camera geometry was informed by coverage calculations and preliminary occlusion testing, which showed that angled auxiliary views improved visibility into partially occluded fruit while preserving overlap with the centre view for registration. External calibration between the D435 RGB stream and each auxiliary RGB camera was performed using a 7×5 asymmetric checkerboard target and OpenCV stereo calibration with a 0.5 px [9] threshold due to its common use in industry.



Figure 1: Multi-view camera coverage of a tomato plant

Table 2: Mechanical User Requirements and Engineering Specifications

User Requirement	Engineering Specification	Final Design Response
Adjustable sensing height for greenhouse tomato plants	0.8–2.2 m height range	Telescoping mast with spring-loaded lock
Stable sensor geometry	No observable drift during normal operation	Bolted tab-and-slot mounts
Greenhouse durability	Enclosure $\geq$ IP54, corrosion-resistant materials	IP54 enclosure, PETG mounts, stainless hardware
Modular iteration and maintenance	Replaceable subassemblies	Independent camera mounts on common mast

C. MACHINE LEARNING DESIGN

Machine learning design was guided by five requirements: detection accuracy above 90% mean average precision for both ripeness and pollination classification, real-time inference suitable for a cloud-hosted microservice pipeline, robustness to the visual variability of greenhouse environments (occlusion, lighting, overlapping fruit), compatibility with low-resolution edge-captured imagery from the Raspberry Pi camera array, and minimal dependency on large labelled datasets specific to Ontario greenhouse conditions.

Model architecture selection – The base detection framework was selected through a weighted decision matrix comparing YOLOv8 Nano, YOLOv5 Small, and Faster R-CNN across inference speed, detection accuracy, model size, ease of deployment, and community support. YOLOv8 Nano was selected for its favourable balance of speed and accuracy at a model size under 15 MB, making it suitable for cloud deployment on free-tier GPU infrastructure. To improve contextual reasoning,

particularly the ability to distinguish visually similar classes such as yellow leaves from yellow flowers, the standard C2f module in the YOLOv8 Nano backbone was replaced with a C3TR (Transformer) block, producing a custom yolov8n-c3tr architecture. The C3TR block introduces a self-attention mechanism that enables the model to leverage global spatial context within each feature map, improving discrimination between fine-grained classes without significantly increasing model size or inference latency.

Tomato Ripeness Model – The Tomato Ripeness Model classifies detected fruit into three stages: Unripe, Half-Ripe, and Ripe. Training data was compiled from three sources: a Greenhouse Tomato dataset (~372 whole-plant images), the LaboroTomato dataset (~804 on-vine images with six original classes remapped to three), and ~30 manually annotated field images oversampled 5× to 150 effective samples. This multi-source strategy addressed domain shift between publicly available close-up crop datasets and the whole-plant, partially occluded views captured by VineVision’s camera array. A conservative fine-tuning learning rate of 0.00005 was used to preserve base model knowledge while adapting to greenhouse-specific conditions. The model was further augmented with an HSV colour-based post-processing classifier applied per detected region: pixels are filtered by saturation ($S > 60$) and value ($V > 60$) to isolate tomato skin, then classified by hue ranges (Red/Ripe: 0–10 and 170–180; Orange/Half-Ripe: 10–25; Green/Unripe: 35–85). A fruit is labelled Ripe if $\geq 35\%$ of qualifying pixels fall in the red range, and Half-Ripe if $\geq 25\%$ are orange or a combined threshold is met. This hybrid approach improved robustness against lighting variation compared to relying on the neural network alone. The final model achieved a mAP@50 of 0.947, surpassing the 90% engineering specification.

Flower Pollination Model – The Flower Pollination Model detects and classifies flowers into three pollination stages: Bud (pre-bloom), Anthesis (actively blooming), and Post-Anthesis (post-bloom). The model was trained on the Tomato-Flower-3-Class dataset from Kaggle (3,636 images, 15,757 annotated instances) using the same custom YOLOv8n-C3TR architecture. Training was conducted on a university HPC cluster with CUDA acceleration for 50 epochs using a batch size of 32, image size of 640×640, AdamW optimizer, and momentum of 0.95. Data augmentation included HSV jitter (to ensure that the model could classify blurry images that were taken on the move), horizontal flip, and mosaic augmentation. Per-class confidence thresholds were tuned during inference (Bud: 0.05, Anthesis: 0.25, Post-Anthesis: 0.25) to balance precision and recall across classes with varying visual distinctiveness. The final model achieved a mAP@50 of 0.902 with precision of 0.832, recall of 0.841, and inference speed of ~9.1 ms per 640×640 image, surpassing the 90% accuracy target.

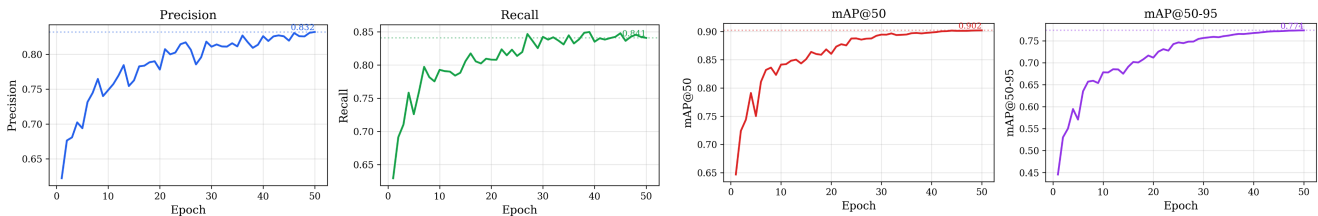


Figure 2: Multi-view camera coverage of a tomato plant

Deployment architecture – Both models are hosted on HuggingFace Spaces using a Gradio and FastAPI server with ZeroGPU (H200) acceleration. The inference server exposes dedicated REST API endpoints for each classification task. A thread-safe model pool with configurable concurrency supports parallel inference requests from the microservice backend. Inference latency averages ~9 ms per image with preprocessing at ~1 ms and postprocessing at ~20 ms, enabling the Classify Service to process a full four-camera scan within the system’s response time requirements.

Table 3: Machine Learning User Requirements and Engineering Specifications

User Requirement	Engineering Specification	Final Design Response
Accurate ripeness classification for yield forecasting	Ripeness model mAP@50 $\geq$ 0.90	YOLOv8n-C3TR + HSV post-processing; mAP@50: 0.947
Accurate pollination stage detection for growth tracking	Pollination model mAP@50 $\geq$ 0.90	YOLOv8n-C3TR trained on 3,636 images; mAP@50: 0.902
Real-time inference compatible with scan workflow	Inference latency $<$ 15 s per image	~30 ms total per image
Robust detection under greenhouse conditions	$\geq$ 80% precision and recall	Precision: 0.832, Recall: 0.841; HSV post-processing
Deployable without on-site GPU hardware	Cloud-hosted inference, no local GPU	HuggingFace Spaces with ZeroGPU; REST API

D. SOFTWARE PLATFORM DESIGN

Through grower consultations, the primary dashboard user requirements were identified as: (i) accessible on any device without installation or IT support, (ii) outputs must be interpretable by a non-technical user without training, and (iii) the system must reliably handle variable load across scanning, classification, and viewing phases of operation.

Monolith vs, Microservice – The primary backend architectural decision was between a monolith and independent microservices. A monolithic deployment would have simplified initial development but introduced resource contention under the system's asymmetric load profile. During a scan, the Raspberry Pi generates high upload volume with minimal dashboard traffic, while during harvest planning the inverse is true. Independent scaling of each component was therefore identified as a hard requirement. This led to a microservice architecture where the Upload Service, Classify Service, and Results Service are decoupled, each able to scale in response to its specific demand, and containerized via Docker behind a central API gateway.

Database Selection – The key database decision was whether to use a relational store such as PostgreSQL or a document store. Each scan session produces variable-length detection records such as fruit counts, ripeness labels, and depth measurements that differ in structure across rows and sessions. A fixed relational schema would require significant restructuring to accommodate this variability. MongoDB was selected for its ability to handle both raw image data and structured classification metadata within a single flexible document schema, reducing integration overhead and simplifying per-session storage.

API Design – The key API design decision was whether to expose a single unified endpoint to both clients or to separate endpoints by client type. A unified approach would have simplified routing but coupled the Pi's upload traffic with the frontend's read requests at the API level, reintroducing the contention the microservice architecture was designed to avoid. The gateway therefore exposes two distinct API surfaces: upload endpoints (/uploadData, /uploadClassify, /demoClassify) exclusively for the Raspberry Pi, and results endpoints (/getSummaryResults, /getDetailedRowData, /getTrends)

exclusively for the React frontend. This separation ensures that high-volume scan uploads do not interfere with dashboard read performance.

Table 4: Software and Dashboard Platform Engineering Specifications

User Requirement	Engineering Specification
Accessible on any device without installation	Web-based React frontend, no client software required
Interpretable by non-technical users	Three dashboard views: summary results, detailed row data, trends
Handle dynamic load across system phase	Independent microservices scaled per component via Docker
Store images and classification metadata together	MongoDB document store with flexible per-session schema

IV. SUMMARY OF SOCIAL, ECONOMIC, AND ENVIRONMENTAL IMPACTS

VineVision's sustainability was evaluated across three pillars using Adjusted GPM for social impact, SDG analysis for economic impact, and GRANTA EcoAudit with TCP Circular Economy analysis for environmental impact.

Social – VineVision's primary contribution is worker protection. Replacing manual crop counting in pesticide-exposed environments with non-invasive camera-based data collection directly reduces occupational health risk, while default face blurring addresses workplace surveillance concerns [10][11]. The main risk is transition management as without structured onboarding, lower-skilled workers may be disproportionately disadvantaged by the shift to technology-dependent workflow [12].

Economic – VineVision delivers approximately 90% accurate yield forecasting on a ~\$500 hardware investment requiring no infrastructure modification, offering strong ROI for growers in a sector generating over \$800 million annually that still relies on manual methods with up to 30% error rates [13][14]. The primary risk is hidden long-term cost: ongoing cloud computing fees inflate true total cost of ownership beyond the hardware price, potentially burdening smaller operations [15].

Environmental – The prototype's aluminum structural components achieve approximately 75% energy reclamation at end of life, and its 20W operational draw earns a perfect use-phase circularity score of 10. The critical liability is the electronic components, which account for over 52% of total system CO₂ emissions with zero reclamation potential, yielding an end-of-life circularity score of 2 out of 10 [16]. Consolidating to a single processing unit in future iterations would reduce e-waste and embodied carbon by an estimated 75% [17]. Over a 3 year lifespan, VineVision is estimated to produce about 75 kg CO₂e. That impact could be offset by preventing roughly 46 kg of tomatoes from being wasted. At the industry level, given the current market size and VineVision's estimated effect on forecasting and waste reduction, the system has the potential to help save up to 23,000 tonnes of tomatoes per year.

V. DESIGNED SOLUTION

VineVision consists of four integrated subsystems: a custom mechanical assembly, an off-the-shelf camera array, and a custom software and ML pipeline.

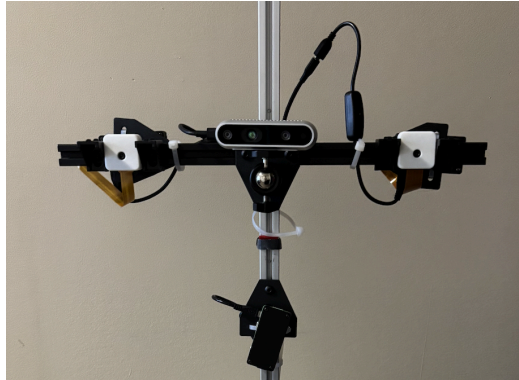


Figure 3: VineVision Prototype

The mechanical assembly is a custom-fabricated adjustable vertical mast mounted on a modified off-the-shelf utility cart. Mast height is repositionable throughout the growth cycle to maintain optimal camera alignment with the fruit zone. An IP54-rated recycled plastic waterproof enclosure houses the compute, power, and communication hardware and is integrated into the cart frame to withstand the greenhouse environment.

The camera system consists of three off-the-shelf Raspberry Pi Camera Module 3 units for RGB image capture and one off-the-shelf Intel RealSense D435 for depth sensing. Custom 3D-printed enclosures were fabricated by the team to mount all 4 cameras at fixed angles on the mast. The RealSense extracts tomato dimensions, estimates volume via ellipsoid modelling, and converts to weight using a research-backed density formula.

The software pipeline is entirely custom-built. Two models were trained and hosted on HuggingFace Hub: a Tomato Ripeness Model (Unripe, Half-Ripe, Ripe; mAP@50: 0.947) and a Flower Pollination Model (Bud, Anthesis, Post-Anthesis; mAP@50: 0.902), both surpassing the 90% accuracy target. Scan data is transmitted to a cloud-hosted microservice backend, processed, and delivered through a custom React dashboard. The system requires no infrastructure modification and delivers zone-level yield predictions up to 8 weeks ahead of harvest.

VI. SUMMARY OF DESIGN EVALUATION AND VALIDATION

VineVision's design evaluation followed an iterative approach with two phases, with each subsystem tested independently before integration-level validation was conducted on the assembled prototype. Hardware, machine learning, and software components were validated against the engineering specifications defined in Section III, with findings from each testing cycle informing refinements to the subsequent iteration.

A. HARDWARE & MECHANICAL SYSTEM

Phase 1 used subsystem-level testing to identify limitations in synchronization, structural rigidity, and field-of-view coverage. Phase 2 verified the final prototype using occlusion testing, power testing, qualitative operational stability checks, and cost verification.

Phase 1 focused on identifying the main hardware and mechanical failure modes of the early prototype. Synchronization testing showed that a USB webcam configuration introduced 50–150 ms inter-camera jitter, producing visible misalignment between views and making it unsuitable for synchronized multi-camera capture. Early motion and loading tests also showed that friction-based mounts and the initial mast configuration did not provide sufficient rigidity for reproducible captures. In parallel, a preliminary occlusion study was used to compare camera arrangements. As shown in Appendix C.1, the centre depth camera alone achieved 71% recall, while the final multi-view arrangement achieved 85%, confirming that a single-view layout did not provide adequate coverage in partially occluded vine conditions. Calibration quality was also verified through stereo reprojection error, with final RMS values of 0.41 px and 0.47 px, confirming that inter-camera alignment was sufficiently accurate for multi-view registration.

Phase 2 evaluated whether the final integrated system satisfied the key hardware and mechanical requirements. Capture and upload performance were verified through repeated stop-and-capture trials on the assembled system. As summarized in Appendix C.2, local capture latency ranged from 0.77 s to 1.05 s, upload latency ranged from 2.1 s to 3.1 s, and total trigger-to-backend latency ranged from 2.9 s to 4.3 s. Average values were 0.82 s for local capture, 2.40 s for upload, and 3.20 s end-to-end, confirming that the final architecture could support a practical workflow. Mechanical stability during normal operation was assessed qualitatively through pushing, turning, stopping, mast adjustment, and manual tilt checks. As seen in Appendix C.3, the system remained stable under normal workload, with no visible sway, no wheel lift, no forward pitch during stopping, and no observable mast slip under payload. Cost verification was also completed through the final bill of materials in Appendix C.4, which documented the prototype hardware components and their associated costs. Together, these results provided preliminary verification that the final hardware and mechanical system met its intended requirements for coverage, power, operational stability, and prototype feasibility.

Further hardware and mechanical validation should focus on long-duration robustness and real deployment conditions. It should include full-shift battery endurance under repeated capture and upload cycles, long-term calibration drift after extended cart motion and repeated mast adjustments, mechanical stability and vibration sensitivity during movement over uneven greenhouse surfaces, and thermal performance of the Raspberry Pi nodes during sustained operation. Additional fault testing under temporary camera, power, or network interruptions would clarify system reliability and recovery behaviour. These tests would strengthen confidence that the prototype not only meets its nominal specifications, but can also maintain performance, stability, and usability during continuous greenhouse deployment.

B. MACHINE LEARNING SYSTEM

Phase 1 established baseline model performance and identified architectural limitations in the early prototype. Phase 2 validated the final models against engineering specifications and confirmed integration with the production microservice pipeline.

Phase 1 focused on training and evaluating initial model candidates to determine whether the custom YOLOv8n-C3TR architecture could meet the 90% mAP@50 target. Early pollination model training

on a local development machine (Apple MPS backend) revealed that the standard YOLOv8 Nano backbone struggled to distinguish between Anthesis and Post-Anthesis classes due to subtle visual differences, motivating the introduction of the C3TR Transformer block for improved contextual reasoning. Initial ripeness classification using the neural network alone showed inconsistent performance under variable greenhouse lighting, with the model frequently misclassifying Half-Ripe tomatoes as either Unripe or Ripe depending on illumination. This finding led to the development of the hybrid HSV colour post-processing classifier as a complementary verification step. An extended 100-epoch training run (mAP@50: 0.891) was also compared against the 50-epoch run (mAP@50: 0.902), confirming that additional training epochs did not improve generalization and that the 50-epoch configuration was optimal. Dataset composition was also evaluated: fine-tuning the ripeness model on greenhouse-specific whole-plant images at a conservative learning rate (0.00005) was necessary to bridge the domain gap between publicly available close-up datasets and VineVision’s multi-camera capture perspective.

Phase 2 validated the final trained models against the defined engineering specifications and evaluated end-to-end integration performance. The Tomato Ripeness Model achieved a mAP@50 of 0.947 and the Flower Pollination Model achieved a mAP@50 of 0.902, both surpassing the 90% target. Pollination model precision reached 0.832 and recall reached 0.841, confirming that the model reliably detects the majority of ground-truth objects while maintaining low false-positive rates. Per-class confusion matrix analysis (Figure D.3) of the Flower Pollination Model on the full 3,636-image validation set revealed that Anthesis achieved the highest per-class mAP@50 of 0.949 with a recall of 0.899, while Bud reached 0.910 mAP@50 with a precision of 0.860. Post-Anthesis was the most challenging class (mAP@50: 0.846, precision: 0.762), primarily due to visual similarity with late-stage Anthesis flowers; however, all three classes individually exceeded the 80% precision and recall targets established in the engineering specifications.

Further machine learning validation should focus on longitudinal performance across a full growing season, as model accuracy may degrade as plant canopy density increases and fruit appearance shifts over time. Testing on tomato varieties beyond the training distribution would quantify generalization limits. Additionally, integration of the Fruit Set Detection model (Bud vs. Fruit classification trained on the TomatoMAP dataset) into the production pipeline would enable earlier yield prediction by detecting fruit formation before ripeness classification becomes applicable, representing a natural extension of the current two-model system.

C. SOFTWARE SYSTEM

Software validation involved two phases. Phase 1 functional testing, a gallery walk demonstration, confirmed core system integration but revealed three major issues: poor dashboard usability for non-technical users, insufficient UI navigation, and excessive ML inference latency (45 seconds, 3x the threshold).

These findings drove an architectural redesign before phase 2. Following the microservice backend transition and dashboard redesign, usability testing was conducted with 8 testers, identifying and resolving 12 defects. End-to-end integration testing across 25 test cases then achieved a 100% pass rate. ML inference latency improved significantly to approximately 4 seconds per image (an 11x improvement). Weight estimation accuracy was validated against manual ground truth measurements across 30 tomatoes. 27 of 30 measurements (90%) were within 5% of the actual weight, meeting the accuracy specification, as shown in Figure D.2.

VII. DESIGN SAFETY AND REGULATIONS

Electrical safety was addressed by housing the Power and Control Unit in an IP54-rated waterproof enclosure, selected in accordance with IEC 60529 to protect against moisture and dust ingress inherent to the greenhouse environment [17]. This directly mitigates the risk of electrical fault or short circuit caused by humidity exposure, protecting both the operator and the equipment. All onboard electronics were designed to comply with CSA Group certification requirements under CSA C22.2, Canada's applicable standard for electrical equipment safety, ensuring the system meets the regulatory threshold for safe deployment in an industrial workplace setting [19].

Mechanical safety was considered in the context of manual operation by a single operator in shared workspace conditions where greenhouse workers are present. The mast assembly includes a locking mechanism to prevent unintended height displacement during operation, eliminating the risk of the mast dropping unexpectedly onto a worker. Cart stability under full mast and camera load was addressed during the mechanical design phase to minimize tip-over risk, consistent with the general duty provisions of Ontario's Occupational Health and Safety Act (OHSA), which requires employers to ensure equipment used in the workplace does not endanger workers [20].

Data privacy obligations were addressed in response to VineVision's camera system operating in an active workplace following Canada's Personal Information Protection and Electronic Documents Act (PIPEDA) [21]. Worker face blurring was implemented directly in the image processing pipeline as a technical control, ensuring no identifiable images of workers are retained in the system or surfaced through the dashboard. Access to scan data stored in MongoDB is restricted to authorized users to prevent unauthorized disclosure of operational data, consistent with PIPEDA's accountability and safeguarding principles.

Risk assessment was conducted using a Risk Matrix framework, evaluating each identified hazard against likelihood and severity to prioritize mitigation efforts. The highest priority risks identified were electrical fault in a humid environment, mechanical instability of the cart under mast load, and unauthorized data access. Each was mitigated through the design controls described above: IP54-rated enclosure, locking mast mechanism, and PIPEDA-compliant face blurring respectively.

VIII. LIMITATIONS OF DESIGNED SOLUTION

Three primary limitations were identified in the current implementation of VineVision. First, the system lacks autonomous positional awareness within the greenhouse, requiring an operator to manually navigate the cart along each row. This dependency on human intervention limits scan frequency and introduces variability in data collection consistency, as coverage is contingent on grower availability rather than a fixed autonomous schedule. Second, the mast and camera positioning were designed specifically around the growth geometry of tomato vines, meaning the system cannot be directly redeployed for other greenhouse crops such as peppers or cucumbers without mechanical redesign of the mast height range and camera angles. Third, the weight estimation pipeline relies on ellipsoid modelling to approximate tomato volume from depth measurements, which assumes regular fruit geometry. Occluded, clustered, or irregularly shaped fruit may produce less accurate volume estimates, introducing compounding uncertainty into the final kilogram-per-row forecast [22].

Two longer-term design impacts were identified as outside the scope of this project but relevant to full commercial deployment. VineVision currently operates as a standalone system with no integration into

existing greenhouse management platforms such as Priva or other ERP systems widely adopted across Ontario's greenhouse sector [23]. Without this integration, growers must manually cross-reference VineVision's yield forecasts with their existing operational workflows, reducing the practical utility of the system in a commercial setting. Additionally, VineVision's current scope is limited to tomato production, which, while representing the largest share of Ontario greenhouse output, excludes peppers and cucumbers that together account for a significant portion of the industry's \$2.5 billion in annual sales [2]. Expanding model training and mechanical design to support multiple crop types would be a necessary step toward broader commercial viability.

IX. CONCLUSIONS AND RECOMMENDATIONS

VineVision successfully demonstrated that affordable, infrastructure-light computer vision can close the yield forecasting gap facing Ontario greenhouse tomato growers. Both ML models surpassed the 90% mAP@50 accuracy target. The Tomato Ripeness Model achieved 0.947 and the Flower Pollination Model achieved 0.902, validating the core technical approach. Weight estimation fell within 5% of ground truth for the majority of tested tomatoes, confirming the viability of the ellipsoid modelling pipeline. The microservice backend architecture proved appropriate for the system's asymmetric load profile, and the cart-mounted design was validated to deploy on existing greenhouse equipment without infrastructure modification. While core technical specifications were met, autonomous positional awareness within the greenhouse and full-season validation against commercial harvest data were not achieved within the scope of this project, and represent the most critical gaps in the current implementation.

Four recommendations are identified for a future group taking over this project. First, autonomous row navigation should be implemented through GPS or RFID-based positioning to eliminate operator dependency during scanning and improve data collection consistency. Second, the ML models and mechanical mast design should be extended to support additional crop types, specifically peppers and cucumbers, to broaden VineVision's addressable market within Ontario's greenhouse sector. Third, VineVision should be integrated with existing greenhouse management platforms such as Priva to embed yield forecasts directly into growers' existing operational workflows, removing the need to cross-reference systems manually. Fourth, a full-season deployment in a live commercial greenhouse should be conducted to validate forecast accuracy against actual harvest data at scale, providing the real-world performance evidence needed to support commercial adoption.

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APPENDIX

Appendix A. Weighted Decision Matrices

Table A.1 Camera Platform Selection Matrix

Criterion	Weight	RPi Camera Module 3	Weighted Score	USB Webcam Array	Weighted Score
Synchronization performance	0.30	5	1.50	2	0.60
Integration complexity	0.20	4	0.80	3	0.60
Cost	0.15	4	0.60	3	0.45
Image quality	0.15	4	0.60	3	0.45
Reliability	0.10	4	0.40	3	0.30
Modularity	0.10	5	0.50	2	0.20
Total	1.00		4.40		2.60

Table A.2 Synchronization Method Selection Matrix

Criterion	Weight	Sequential Trigger	Weighted Score	Shared-Timestamp Software Trigger	Weighted Score	Hardware Trigger	Weighted Score
Timing performance	0.30	2	0.60	4	1.20	5	1.50
Wiring complexity	0.25	5	1.25	4	1.00	1	0.25
Implementation simplicity	0.20	5	1.00	4	0.80	2	0.40
Reliability	0.15	2	0.30	4	0.60	5	0.75
Scalability	0.10	2	0.20	4	0.40	4	0.40
Total	1.00		3.35		4.00		3.30

Appendix B. Design Calculations

Eq B.1 Side-camera coverage estimate

At a nominal standoff distance of 0.5 m and a side-view angle of 30° , the lateral coverage increase is calculated as follows:

$$x = d \tan(\theta) = 0.5 \tan(30^\circ) \approx 0.289 \text{ m}$$

Each side camera therefore increased lateral coverage by approximately 0.29 m relative to the centre view. This geometry supported the implementation of $\pm 30^\circ$ side cameras for effective occlusion reduction while maintaining the overlap region required for multi-view registration and calibration.

Eq B.2 Power budget used for sizing

The total operational power draw (P_{total}) is derived from the summation of the power requirements for the Raspberry Pi 4 (P_{Pi4}), three Raspberry Pi Zero nodes (P_{3xZero}), the RealSense sensor (P_{D435}), and auxiliary peripherals ($P_{peripherals}$): $P_{total} = 5.1 + 2.7 + 3.5 + (2.1 \text{ to } 4.1) = 13.4 \text{ to } 15.4 \text{ W}$

Appendix C. Verification Tables

Table C.1 Occlusion Test Coverage

Configuration	Recall
Centre depth camera only	71%
Multi-view arrangement	85%
4 cameras vertically stacked	79%

Recall was calculated as detected visible fruit / manually labeled ground-truth fruit across N representative greenhouse scenes.

Table C.2 Qualitative Operational Stability Assessment

Test Condition	Observation
Straight pushing on flat surface	Stable, no visible sway
Turning at normal walking speed	Stable, no wheel lift
Stopping from motion	Stable, no forward pitch
Mast adjustment under payload	No observable slip
Manual lateral tilt	Stable under normal workload

Table C.3 VineVision BOM

Item	Quantity	Unit Cost (CAD)	Subtotal (CAD)	Notes
Raspberry Pi 4 Model B	1	95	95	Central coordination and backend communication
Raspberry Pi Zero 2 W	3	24	72	One per auxiliary camera node
Intel RealSense D435	1	180	180	Central RGB-depth sensing
Raspberry Pi Camera Module 3	3	25	75	Auxiliary RGB cameras
2020 aluminum extrusion / mast structure	1 lot	18	18	Mast and mounting frame
Utility cart / base modifications	1 lot	10	10	Prototype base allocation
IP54 enclosure	1	12	12	Electronics protection
Cable glands, wiring, connectors	1 lot	8	8	Sealing and interconnects
PETG for printed mounts	1 lot	6	6	Camera brackets and holders
Stainless steel fasteners and hardware	1 lot	4	4	Corrosion-resistant assembly

Battery mount / power integration hardware	1 lot	5	5	Mounting and distribution hardware
Total			485	

Appendix D. Verification Plots

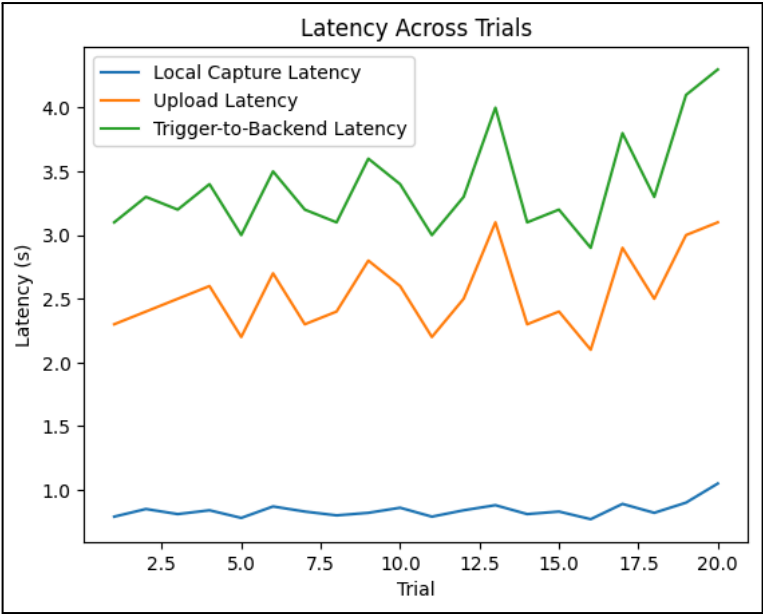


Figure D.1 Capture and Upload Latency Trials Plot

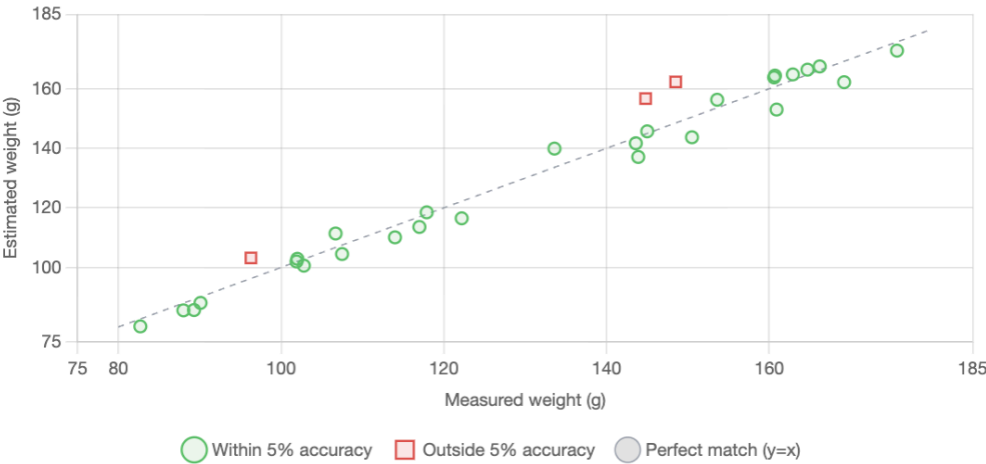




Figure D.2 Weight Estimation Trials

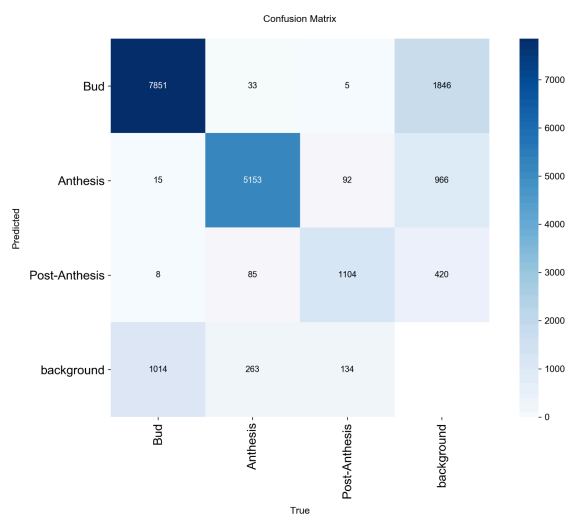


Figure D.3 Confusion Matrix for Tomato Ripeness Model